



**B.Sc. (General/Special) Degree
Programme
Faculty of Applied Sciences
University of Sri Jayewardenepura**

ICT 1022 2.0 Computer System Architecture

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SOFTWARE AND ARCHITECTURE TYPES



Introduction

A software or a program consists of a set of instructions required to solve a specific problem.

- A program to sort a set of numbers.
- A program to find the roots of a quadratic equation.
- A program to find the inverse of a matrix.
- The C compiler that translates a C program to machine language.
- The editor program that helps us in creating a document.
- The operating system that helps us in using the computer system.



Types of Programs

Broadly we can classify programs/software into two types:

a) Application Software

- Which helps the user to solve a particular user-level problem.
- May need system software for execution.

b) System Software

- A collection of programs that helps the users to create, analyze and run their programs.



(a) Application Software

Application software helps users solve particular problems.

- In most cases, application software resides on the computer's hard disk or removable storage media (DVD, USB drive, etc.).
- Typical Examples:
 - Financial accounting package
 - Mathematical packages like MATLAB or MATHEMATICA – An app to book a cab
 - An app to monitor the health of a person



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(b) System Software

- System software is a collection of programs, which helps users run other programs.
- Typical operations carried out by system software:
 - Handling user requests
 - Managing application programs and storing them as files
 - File management in secondary storage devices
 - Running standard applications such as word processor, internet browser, etc.
 - Managing I/O units
 - Program translation from source form to object form



Linking and running user programs

Some very commonly used system software:

- Operating system (WINDOWS, LINUX, MAC/OS, ANDROID, etc.)
 - Instance of a program that never terminates.
 - The program continues running until either the machine is switched off or the user manually shuts down the machine.
- Compilers and assemblers
- Linkers and loaders
- Editors and debuggers

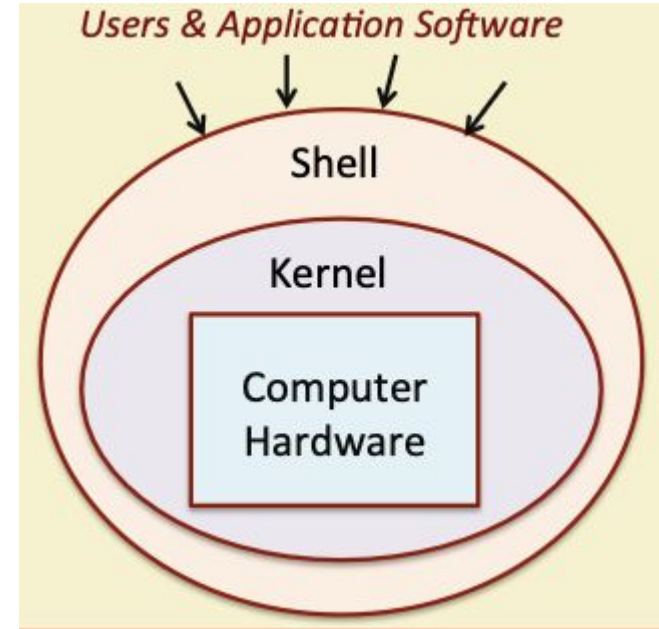


Operating System

Provides an interface between computer hardware and users.

Two layers:

- a) Kernel: contains low-level routines for resource
- b) Shell: provides an interface for the users to interact with the computer hardware through the kernel



Depending on the intended use of the computer system, the goal of the OS may differ.

- Classical multi-programming systems
 - Several user programs loaded in memory.
 - Switch to another program when one program gets blocked due to I/O.
 - Objective is to maximize resource utilization.
- Modern time-sharing systems
 - Widely used because every user can now afford to have a separate terminal.
 - Processor time shared among a number of interactive users.

Objective is to reduce the user response time.



- Real-time systems
 - Several applications are running with specific deadlines.
 - Deadlines can be either hard or soft.
 - Interrupt-driven operation
- processor interrupted when a task arrives.
 - Examples: missile control system, industrial manufacturing plant, patient health monitoring and control system, automotive control system, etc.
- Mobile (phone) systems
 - Here user responsiveness is the most important.
 - Sometimes a program that makes the system slow or hogs too much memory may be forcibly stopped.



Classification of Computer Architecture

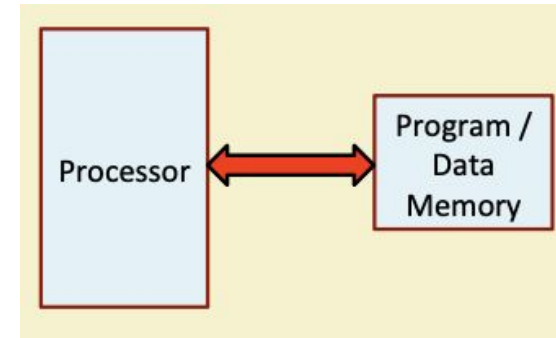
- Broadly can be classified into two types:
 - a) Von-Neumann architecture
 - b) Harvard architecture
- How is a computer different from a calculator?
 - They have similar circuitry inside (e.g. for doing arithmetic).



von-Neumann Architecture

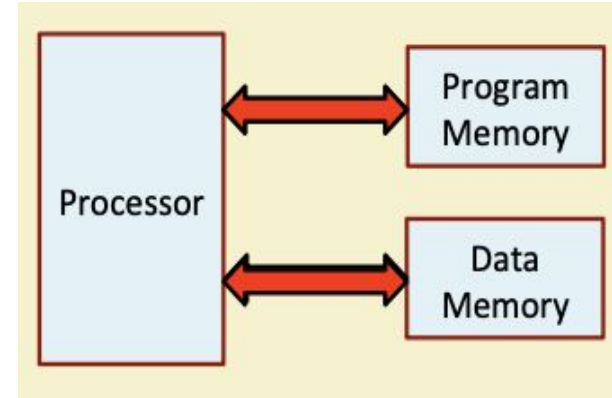
Instructions and data are stored in the same memory module.

- More flexible and easier to implement.
- Suitable for most of the general purpose processors.
- General Disadvantage:
 - The processor-memory bus acts as the bottleneck.
 - All instructions and data are moved back and forth through the pipe.



Harvard Architecture

- Separate memory for program and data.
 - Instructions are stored in program memory and data are stored in data memory.
- Instruction and data accesses can be done in parallel.
- Some micro controllers and pipelines with separate instruction and data caches follow this concept.
- The processor-memory bottleneck remains.



Emerging Architectures

Several architectural concepts have been proposed that deviate significantly from the von-Neumann or Harvard model.

- There is a proposal for in-memory computing architecture, where storage and computation can be done in the same functional unit.
 - Memristors are projected to make this possible in the near future.
 - Memristors can be used in high capacity non-volatile resistive memory systems.
 - Memristors within the memory can also be controlled to carry out some computations.



Pipeline in Executing Instructions

- Instruction execution is typically divided into 5 stages:
 - Instruction Fetch (IF)
 - Instruction Decode (ID)
 - ALU operation (EX)
 - Memory Access(MEM)
 - Write Back result to register file(WB)
- These five stage can be executed in an overlapped fashion in a pipeline architecture.
 - Results in significant speedup by overlapping instruction execution.



Basic 5-stage Pipelining Diagram

Instruction	Pipeline Stage						
	IF	ID	EX	MEM	WB		
1	IF	ID	EX	MEM	WB		
2		IF	ID	EX	MEM	WB	
3			IF	ID	EX	MEM	WB
4				IF	ID	EX	MEM
5					IF	ID	EX
Clock Cycle	1	2	3	4	5	6	7



How can Harvard Architecture Help?

In clock cycle 4, instruction 4 is trying to fetch an instruction (IF), while instruction 1 may be trying to access data (MEM).

- In von-Neumann architecture, one of these two operations will have to wait resulting in pipeline slowdown.
- In Harvard architecture, the operations can go on without any speed penalty as the instruction and data memories are separate.



Next

Instruction Set Architecture



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Thank
you!

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